

Signal Processing

Exercise 3 – Fourier transform

Summer Term 2026

General

The Fourier determines the (complex) harmonics from which a signal $x(t)$ can be synthesized (at least theoretically). The Fourier transform consists of a (frequency dependent) amplitude and (frequency dependent) phase. The Fourier transform therefore “measures” the “frequency content” of a signal in a certain manner. This “frequency content” is called the spectrum of the signal.

Transformation equation

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) \cdot e^{-j\omega t} dt$$

$$X(f) = \int_{-\infty}^{+\infty} x(t) \cdot e^{-j2\pi f t} dt$$

Inverse transformation equation

$$x(t) = \frac{1}{2\pi} \cdot \int_{-\infty}^{+\infty} X(j\omega) \cdot e^{j\omega t} d\omega$$

$$x(t) = \int_{-\infty}^{+\infty} X(f) \cdot e^{j2\pi f t} df$$

Difference between the Fourier- and Laplace transform

Fourier	Laplace
$X(j\omega) = \int_{-\infty}^{+\infty} x(t) \cdot e^{-j\omega t} dt$	$X(s) = \int_{-\infty}^{+\infty} x(t) \cdot e^{-st} dt, \quad s = \sigma + j\omega$
→ Analysis of the frequency spectrum	→ Analysis of the stability → Analysis of the transient response

The Fourier transform is a special form of the Laplace transform ($X(j\omega) = X(s)|_{s=j\omega}$)

Existence of the Fourier transform

The Fourier transform exists if the following applies: $|X(j\omega)| < \infty$

→ Time-limited signals with limited amplitude always fulfill this condition.

Task 1 – Fourier transform of a rectangular signal

- a) Determine the Fourier transform $X(j\omega)$ of the time-continuous signal

$$x(t) = \text{rect}\left(\frac{t}{2T_0}\right)$$

by using the transformation equation of the Fourier transform.

- b) Discuss the influence of the variable T_0 on the spectrum of the signal $x(t)$.

Task 2 – Fourier transform of the dirac impulse (Bonus task)

Determine the Fourier transform $X(j\omega)$ of the time-continuous signal

$$x(t) = \delta(t)$$

by using the transformation equation of the Fourier transform.

Task 3 – Fourier transform of a decreasing exponential function (Bonus task)

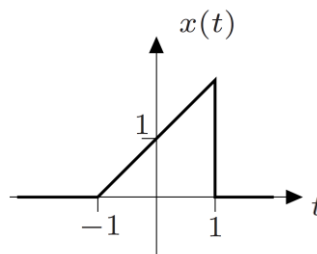
Determine the Fourier transform $X(f)$ of the time-continuous signal

$$x(t) = \sigma(t) \cdot U_0 \cdot e^{-\frac{1}{RC}t}$$

by using the transformation equation of the Fourier transform.

Task 4 – Fourier transformation of a sawtooth-signal (Bonus task)

Given the following signal $x(t)$:



- Determine an equation for the signal $x(t)$.
- Calculate the spectrum $X(j\omega)$ of the signal $x(t)$ using the transformation equation of the Fourier transform..

Task 5 – Inverse Fourier transform of a rectangular signal

Calculate the corresponding time function $x(t)$ of the spectrum $X(f)$

$$X(f) = \text{rect}(f)$$

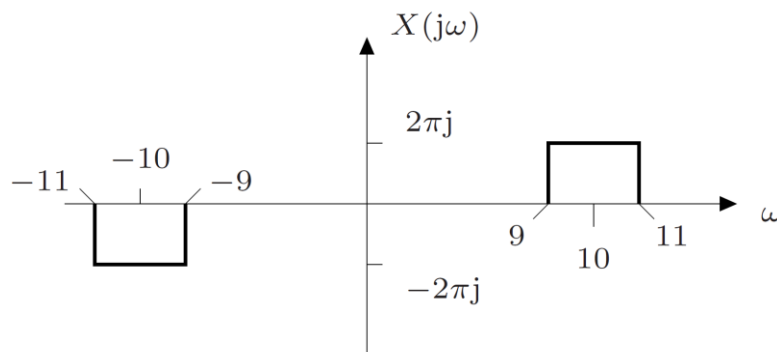
by solving the integral of the inverse Fourier transform.

Task 6 – Inverse Fourier transform

- a) Familiarize yourself with the theorems and correspondences of the Fourier transform.
- b) Calculate $x(t)$ using the correspondence table. First perform a transformation into the frequency domain. Solve the equation in the frequency domain. Then perform an inverse transformation into the time domain.

- I. $x(t) = \text{rect}(t) * \text{rect}(t)$
- II. $x(t) = \sin(\pi t) * \delta\left(t - \frac{1}{2}\right)$

- c) A signal $x(t)$ has the given spectrum:



- I. Calculate the corresponding time signal $x(t)$ via the inverse Fourier transform.
- II. Determine $x(t)$ using the multiplication theorem.

Task 7 – Time and Frequency domain

The magnitude $|Y(j\omega)|$ and phase $\varphi(j\omega)$ of several Fourier transforms (1 – 6) and their corresponding time functions $y(t)$ (A – F), which are not in correct order, are shown on the next page.

$$Y(j\omega) = |Y(j\omega)|e^{j\varphi(j\omega)} \quad \bullet \text{---} \circ \quad y(t)$$

The time functions are characterized by their real part (—) and their imaginary part (- - -).

$$t_1 = \frac{\pi}{\omega_g}.$$

Assign the time function to each corresponding Fourier transform by giving a system theoretic justification.

